Sahaj Bhadja

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Summary

Java Backend Engineer with 2+ years of experience building **microservices**, **production data systems**, and AI-enabled workflows through clean design, testing, CI/CD, observability, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Operating Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **fault-tolerant orchestration** with explicit job-state transitions, failure isolation, recoverable background processing, and observable concurrent workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **production data pipelines**, implementing incremental processing, validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ grading consistency across 10K+ submissions through **structured model evaluation**, model switching, fallback paths, persistent results, and instructor review controls supporting reproducible verification reliably at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **microservice platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time state synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

Skills

Languages: Java, Python, SQL, TypeScript, JavaScript, C++, Bash
Java & Backend: Spring Boot, Object-Oriented Design, Microservices, REST APIs, API Design, FastAPI, Flask, Node.js
AI & Data: LLMs, Retrieval-Augmented Generation, Vector Databases, FAISS, Pinecone, Transformers, PyTorch, Model Evaluation
Cloud & Databases: PostgreSQL, AWS RDS, AWS S3, AWS Glue, Azure, GCP, Docker, Kubernetes
Quality & Operations: Git, GitHub, Jenkins, CI/CD, Unit Testing, Integration Testing, PyTest, Mockito, Observability, Performance Optimization

Projects

Supply Chain GenAI Support Service - Design Concept

- Designed a **Java/Spring Boot GenAI service** concept integrating an LLM API, retrieval, structured prompts, authorization, audit logs, evaluation datasets, grounded responses, human escalation, and operational health metrics.

SaaS Incident and Performance Harness - Design Concept

- Designed a **Java performance harness** concept exercising service endpoints under repeatable workloads, tracking latency, throughput, errors, and resource consumption, preserving benchmark baselines, and surfacing broken-build regressions through CI.

Big-Brain | Live

- Built a **retrieval-augmented generation platform** using LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, and REST APIs, sustaining 95% query accuracy while enabling real-time document analysis reliably for 200+ users.